# Experiment - 12

Student Name: Aditya Pratap Singh
Branch: CSE - AIML
Semester: 5
Subject Name: ADBMS

UID: 24BAI70438
Section/Group: 24AIT_KRG-1_B
Date of Performance: 21/9/26
Subject Code: 24CSP-310

(A)
Design a trigger which whenever there is an insertion on student table then, the currently inserted or deleted row should be printed as it as on the output console window.

Solution:

```
CREATE OR REPLACE FUNCTION FN_STUDENT_AUDIT()
RETURNS TRIGGER
LANGUAGE PLPGSQL
AS
$$
BEGIN
        IF TG_OP = 'INSERT' THEN
        RAISE NOTICE 'INSERTED ROW -> ID: %, NAME: %, AGE: %, CLASS:
%',
         NEW.ID, NEW.NAME, NEW.AGE, NEW.CLASS;
        RETURN NEW;

        ELSIF TG_OP = 'DELETE' THEN
         RAISE NOTICE 'DELETED ROW -> ID: %, NAME: %, AGE: %, CLASS:
%',
        OLD.ID, OLD.NAME, OLD.AGE, OLD.CLASS;
        RETURN OLD;

        END IF;
        RETURN NULL;
END;
```

```
$$

CREATE TRIGGER TRG_STUDENT_AUDIT
AFTER INSERT OR DELETE
ON STUDENT
FOR EACH ROW
EXECUTE FUNCTION FN_STUDENT_AUDIT();

INSERT INTO STUDENT(NAME, AGE, CLASS) VALUES('JOHN', 17, 11);

DELETE FROM STUDENT
WHERE NAME='JACK';
```

(B)
Design a POSTGRESQL triggers that:
Whenever a new employee is inserted in tbl_employee, a record should be added to tbl_employee_audit like:
"Employee name <emp_name> has been added at <current_time>"
Whenever an employee is deleted from tbl_employee, a record should be added to tbl_employee_audit like:
"Employee name <emp_name> has been deleted at <current_time>"

The solution must use PostgreSQL triggers.

Solution:

```
CREATE OR REPLACE FUNCTION AUDIT_EMPLOYEE_CHANGES()
RETURNS TRIGGER
LANGUAGE PLPGSQL
AS
$$
BEGIN
   IF TG_OP = 'INSERT' THEN
      INSERT INTO TBL_EMPLOYEE_AUDIT(message)
      VALUES ('Employee name ' || NEW.EMP_NAME || ' has been added at ' ||
NOW());
      RETURN NEW;
```

```
    ELSIF TG_OP = 'DELETE' THEN
        INSERT INTO TBL_EMPLOYEE_AUDIT(message)
        VALUES ('Employee name ' || OLD.EMP_NAME || ' has been deleted at ' ||
NOW());
        RETURN OLD;
    END IF;

    RETURN NULL;
END;
$$

CREATE TRIGGER TRG_EMPLOYEE_AUDIT
AFTER INSERT OR DELETE
ON TBL_EMPLOYEE
FOR EACH ROW
EXECUTE FUNCTION AUDIT_EMPLOYEE_CHANGES();


INSERT INTO TBL_EMPLOYEE(EMP_NAME, EMP_SALARY) VALUES
('ALAN', 50000);

DELETE FROM TBL_EMPLOYEE
WHERE EMP_NAME='ALAN';

SELECT * FROM TBL_EMPLOYEE_AUDIT;
```